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ICS 421

Implementation of the TODOs

a. Descriptions for Your Implementations

TODO 1: Implementing the Execute Function in SelectNode:execute

Objective:
To complete the execution logic for the SelectPipe in the SelectNode class.

Steps:

1. **Create the Operator**: Instantiate a SelectPipe object.
2. **Run the Operator**: Call the Run method on the SelectPipe, passing in the appropriate pipes and conditions.
3. **Assign Executor**: Set the executor member variable to the recently created SelectPipe object.
4. **Wait for Completion**: Use WaitUntilDone() to ensure that the operation completes before proceeding.

TODO 2: Implementing the ProjectNode Constructor

Objective:
To complete the constructor of the ProjectNode class, ensuring it correctly captures and stores the attributes specified in atts.

Steps:

1. **Iterate Over Attributes**: Loop through the provided NameList of atts variable.
2. **Find Attribute Index**: Use input_schema->Find(atts->name) to locate the index of the attribute in the input schema.
3. **Error Handling**: If the attribute is not found, throw an error.
4. **Store Information**:
    - Save the index to the keepMe[numAttsOutput] array.
    - Store keep_atts[numAttsOutput] array based on the found index for each name and myType.
5. **Update Counter**: Increment numAttsOutput for each valid attribute processed.
6. **Moving to next atts**: go on the next atts if available.

b. Bugs or Difficulties Encountered

1. I encountered a bug while compiling the project in the Comparison.cc file, as shown in the error message below. To resolve this issue, I added the **-std=c++20** flag to the Makefile.

```
thangnguyen@Thangs-MacBook-Pro P_final % make clean
rm -f *.o
rm -f *.out
rm -f y.tab.c
rm -f lex.yy.c
rm -f y.tab.h
rm -f *.stat
rm -f *.tmp
thangnguyen@Thangs-MacBook-Pro P_final % make
g++ -O2 -Wno-deprecated -g -c Util.cc
g++ -O2 -Wno-deprecated -g -c Record.cc
g++ -O2 -Wno-deprecated -g -c Comparison.cc
Comparison.cc:158:14: error: expected expression
                        atts[i] = {"Int", Int};
                                  ^
Comparison.cc:161:14: error: expected expression
                        atts[i] = {"Double", Double};
                                  ^
Comparison.cc:164:14: error: expected expression
                        atts[i] = {"String", String};
                                  ^
3 errors generated.
make: *** [Comparison.o] Error 1
thangnguyen@Thangs-MacBook-Pro P_final %
```

2. I found a bug while compiling the project in the BigQ.cc file, as shown in the error message below. To resolve it, I created a function to cast BigQ to a pointer type and then called the TPMMS function afterward.

```
                                  ^
BigQ.cc:35:36: error: cannot cast from type 'void (BigQ::*)()' to pointer type 'THREADFUNCPTR' (aka 'void *(*)(void *)')
    pthread_create (&worker, NULL, (THREADFUNCPTR) &BigQ::TPMMS, this);
                                   ^~~~~~~~~~~~~~~~~~~~~~~~~~~~
1 warning and 1 error generated.
make: *** [BigQ.o] Error 1
thangnguyen@Thangs-MacBook-Pro P_final %
```

Solution:

```
static void *threadFunc(void *context)
{
    ((BigQ *)context)->TPMMS();
    return nullptr;
}
```

c. And any tips or notes for users/developers to pay attention to.

To run ./test.out [1-3], you must create a directory named data containing all the binary heap files. Additionally, ensure that there is a corresponding file with the .cache extension in the same directory.

d. Running test

Test 1:

```
thangnguyen@Thangs-MacBook-Pro P_final % ./test.out 1
SQL query ----------------

SELECT s.s_name
FROM supplier AS s
WHERE (s.s_suppkey < 100);

----------------------------
1: LEX_ERROR: invalid character at
2: LEX_ERROR: invalid character at
3: LEX_ERROR: invalid character at
4: LEX_ERROR: invalid character at
Optimal predicate (cost 0): ( s_suppkey < 100 )

Query plan ----------------
Write Out(1, 2, -1, -1)
Project(-1, 1, -1, 2)
SelectPipe(-1, 0, -1, 1)
SelectFile(0, -1, -1, 0)
---------------------------------------

Pipe information ------------------------
PipeID 0, Addr: 0x7ff7a6806cc0
PipeID 1, Addr: 0x7ff7a6806d80
PipeID 2, Addr: 0x7ff7a6806e70
-----------------------------------------

[SelectFile] (To pipe 0x7ff7a6806cc0) The number of records from file '../data/bin/supplier.bin.meta': 100
[SelectPipe] The total number of records from pipe 0x7ff7a6806cc0: 100, 99 passed and output to pipe 0x7ff7a6806d80
[Project] Input pipe 0x7ff7a6806d80, output pipe 0x7ff7a6806e70

Results -------------------
s_name: [Supplier#000000001]
s_name: [Supplier#000000002]
s_name: [Supplier#000000003]
s_name: [Supplier#000000004]
s_name: [Supplier#000000005]
s_name: [Supplier#000000006]
s_name: [Supplier#000000007]
s_name: [Supplier#000000008]
s_name: [Supplier#000000009]
s_name: [Supplier#000000010]
s_name: [Supplier#000000011]
s_name: [Supplier#000000012]
s_name: [Supplier#000000013]
s_name: [Supplier#000000014]
s_name: [Supplier#000000015]
s_name: [Supplier#000000016]
s_name: [Supplier#000000017]
s_name: [Supplier#000000018]
s_name: [Supplier#000000019]
s_name: [Supplier#000000020]
s_name: [Supplier#000000021]
s_name: [Supplier#000000022]
s_name: [Supplier#000000023]
s_name: [Supplier#000000024]
s_name: [Supplier#000000025]
s_name: [Supplier#000000026]
s_name: [Supplier#000000027]
s_name: [Supplier#000000028]
s_name: [Supplier#000000029]
s_name: [Supplier#000000030]
s_name: [Supplier#000000031]
s_name: [Supplier#000000032]
s_name: [Supplier#000000033]
s_name: [Supplier#000000034]
s_name: [Supplier#000000035]
s_name: [Supplier#000000036]
                                                    Ln 868, Col 15    Tab Size: 4
```

```
thangnguyen@Thangs-MacBook-Pro P_final % ./test.out 2
SQL query ------------------

SELECT SUM (s.s_acctbal)
FROM supplier AS s
WHERE (s.s_suppkey < 100);

----------------------------
1: LEX_ERROR: invalid character at
2: LEX_ERROR: invalid character at
3: LEX_ERROR: invalid character at
4: LEX_ERROR: invalid character at
Optimal predicate (cost 0): ( s_suppkey < 100 )

Query plan ----------------
Write Out(1, 2, -1, -1)
Sum(-1, 1, -1, 2)
SelectPipe(-1, 0, -1, 1)
SelectFile(0, -1, -1, 0)
----------------------------------------

Pipe information -----------------------
PipeID 0, Addr: 0x7f8bf63040b0
PipeID 1, Addr: 0x7f8bf63041a0
PipeID 2, Addr: 0x7f8bf6304290
----------------------------------------

[SelectFile] (To pipe 0x7f8bf63040b0) The number of records from file '../data/bin/supplier.bin.meta': 100
[SelectPipe] The total number of records from pipe 0x7f8bf63040b0: 100, 99 passed and output to pipe 0x7f8bf63041a0
[Sum] Calculate sum on 99 records from pipe 0x7f8bf63041a0
Sum: 397738.300000

Results -------------------
sum: [397738]
----------------------------
The number of results: 1 rows
thangnguyen@Thangs-MacBook-Pro P_final %
```

Test 3:

```
thangnguyen@Thangs-MacBook-Pro P_final % ./test.out 3
SQL query ------------------

SELECT s.s_name, s.s_acctbal
FROM supplier AS s
WHERE (s.s_suppkey < 100) AND (s.s_acctbal > 1000.0);

----------------------------
1: LEX_ERROR: invalid character at
2: LEX_ERROR: invalid character at
3: LEX_ERROR: invalid character at
4: LEX_ERROR: invalid character at
Optimal predicate (cost 3333.33): ( s_suppkey < 100 ) AND ( s_acctbal > 1000.0 )

Query plan ----------------
Write Out(1, 3, -1, -1)
Project(-1, 2, -1, 3)
SelectPipe(-1, 1, -1, 2)
SelectPipe(-1, 0, -1, 1)
SelectFile(0, -1, -1, 0)
----------------------------------------

Pipe information -----------------------
PipeID 0, Addr: 0x7fdb5a2046c0
PipeID 1, Addr: 0x7fdb5a204780
PipeID 2, Addr: 0x7fdb5a204840
PipeID 3, Addr: 0x7fdb5a204900
----------------------------------------

[SelectFile] (To pipe 0x7fdb5a2046c0) The number of records from file '../data/bin/supplier.bin.meta': 100
[SelectPipe] The total number of records from pipe 0x7fdb5a2046c0: 100, 99 passed and output to pipe 0x7fdb5a204780
[SelectPipe] The total number of records from pipe 0x7fdb5a204780: 99, 79 passed and output to pipe 0x7fdb5a204840

Results -------------------[Project] Input pipe 0x7fdb5a204840, output pipe 0x7fdb5a204900

s_acctbal: [5755.94], s_name: [Supplier#000000001]
s_acctbal: [4032.68], s_name: [Supplier#000000002]
s_acctbal: [4192.4], s_name: [Supplier#000000003]
s_acctbal: [4641.08], s_name: [Supplier#000000004]
s_acctbal: [1365.79], s_name: [Supplier#000000006]
s_acctbal: [6820.35], s_name: [Supplier#000000007]
s_acctbal: [7627.85], s_name: [Supplier#000000008]
s_acctbal: [5302.37], s_name: [Supplier#000000009]
s_acctbal: [3891.91], s_name: [Supplier#000000010]
s_acctbal: [3393.08], s_name: [Supplier#000000011]
s_acctbal: [1432.69], s_name: [Supplier#000000012]
s_acctbal: [9107.22], s_name: [Supplier#000000013]
s_acctbal: [9189.82], s_name: [Supplier#000000014]
s_acctbal: [2972.26], s_name: [Supplier#000000016]
s_acctbal: [1687.81], s_name: [Supplier#000000017]
s_acctbal: [7040.82], s_name: [Supplier#000000018]
s_acctbal: [6150.38], s_name: [Supplier#000000019]
s_acctbal: [9365.8], s_name: [Supplier#000000021]
s_acctbal: [5926.41], s_name: [Supplier#000000023]
s_acctbal: [9170.71], s_name: [Supplier#000000024]
s_acctbal: [9198.31], s_name: [Supplier#000000025]
s_acctbal: [1887.62], s_name: [Supplier#000000027]
s_acctbal: [8080.14], s_name: [Supplier#000000030]
s_acctbal: [5916.91], s_name: [Supplier#000000031]
s_acctbal: [3556.47], s_name: [Supplier#000000032]
s_acctbal: [8564.12], s_name: [Supplier#000000033]
s_acctbal: [4381.41], s_name: [Supplier#000000035]
s_acctbal: [2371.51], s_name: [Supplier#000000036]
s_acctbal: [3017.47], s_name: [Supplier#000000037]
s_acctbal: [2512.41], s_name: [Supplier#000000038]
s_acctbal: [6115.65], s_name: [Supplier#000000039]
s_acctbal: [6942.67], s_name: [Supplier#000000041]
s_acctbal: [6565.11], s_name: [Supplier#000000042]
```